Project Work Explainable AI: Comparative Study of Post-Hoc Methods vs Intrinsic Interpretability in Vision Models

Fouepe Dylan
University of Florence
dylan.fouepe@edu.unifi.it

***Abstract*—As deep learning models become increasingly complex, understanding their decision-making processes is critical. This project presents a comprehensive comparative study of Explainable AI (XAI) methods on visual classification tasks. We evaluate popular post-hoc explainability techniques (SHAP, LIME, GradCAM) against the intrinsic interpretability of Prototypical Part Network (ProtoPNet) architectures. Using two distinct datasets (CLE4EVR and MNMath), we propose a unified evaluation pipeline that measures explanation infidelity, complexity, and localization accuracy. Our findings reveal a striking negative correlation between post-hoc attributions and the internal spatial reasoning of ProtoPNet. While methods like SHAP isolate fine-grained edges, the inherently interpretable ProtoPNet captures broader, context-aware spatial patterns, confirming the theoretical argument that post-hoc methods often misrepresent the true cognitive processes of modern neural networks [1].**

***Index Terms*—Explainable AI, XAI, Deep Learning, ProtoPNet, SHAP, LIME, GradCAM, Computer Vision**

I. Introduction

The rapid advancement of deep neural networks in computer vision has led to unprecedented accuracy but at the cost of transparency. Convolutional Neural Networks (CNNs) like ResNet [2] and Vision Transformers (ViTs) [3] are widely considered “black boxes.” To mitigate this, post-hoc Explainable AI (XAI) methods such as SHAP [4], LIME [5], and GradCAM [6] have been developed to highlight important input features.

However, post-hoc methods have been criticized for their lack of fidelity and potential to generate misleading explanations [7], [8]. As an alternative, intrinsically interpretable models like the Prototypical Part Network (ProtoPNet) [9] incorporate interpretability directly into the architecture by dissecting images into prototypes.

This study systematically compares post-hoc XAI methods against intrinsic interpretability. We pose the following research questions:

- How do SHAP, LIME, and GradCAM perform on standard architectures (ResNet50, ViT) across datasets of varying object scale?
- Does LIME’s reliance on superpixels hinder its localization capability compared to perturbation-based methods?
- Do post-hoc explanations agree with the internal reasoning of an intrinsically interpretable model?

II. Methodology

A. Datasets

We utilized two custom visual datasets with varying characteristics:

- **CLE4EVR**: A dataset composed of 3D objects (spheres, cubes, cylinders) rendered with diverse colors and materials (inspired by CLEVR-Hans [10]). It tests the models’ ability to localize large, well-defined objects in a 3D scene.
- **MNMath**: A dataset containing handwritten mathematical equations. It challenges the models with fine-grained, high-contrast, stroke-based features.

B. Architectures

Three primary architectures were evaluated:

1) **ResNet50**: A standard CNN backbone.
2) **Vision Transformer (ViT)**: A self-attention-based architecture.
3) **ProtoPNet**: An inherently interpretable network built on a ResNet backbone that classifies images based on the L^2 distance between latent features and learned prototypes.

C. Explainability Methods

The comparative pipeline evaluates three post-hoc methods:

- **SHAP**: Computes Shapley values based on expected gradients. The importance ϕ_i of feature i is defined as:

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n - |S| - 1)!}{n!} \times [v(S \cup \{i\}) - v(S)] \quad (1)$$

- **LIME**: Perturbs the image by hiding superpixels and fits a linear surrogate model $g \in G$ to locally approximate the complex model f. The objective is to minimize the fidelity loss $\mathcal{L}$ weighted by proximity π_x, penalized by complexity $\Omega(g)$:

$$\xi(x) = \underset{g \in G}{\arg\min}\ \mathcal{L}(f, g, \pi_x) + \Omega(g) \quad (2)$$

- **GradCAM**: Uses the gradients of the target concept y^c flowing into the final convolutional layer feature maps

A^k. The neuron importance weights α_k^c and the final heatmap $L^c_{Grad-CAM}$ are computed as:

$$\alpha_k^c = \frac{1}{Z}\sum_i \sum_j \frac{\partial y^c}{\partial A^k_{ij}}$$
$$L^c_{Grad-CAM} = ReLU\left(\sum_k \alpha_k^c A^k\right) \tag{3}$$

For ProtoPNet, we extract its **Internal Spatial Heatmap**. The similarity $sim(\mathbf{z}, \mathbf{p}_j)$ between a latent convolutional patch $\mathbf{z}$ and a learned prototype $\mathbf{p}_j$ is computed via inverted L^2 distance:

$$sim(\mathbf{z}, \mathbf{p}_j) = \log\left(\frac{d^2(\mathbf{z}, \mathbf{p}_j) + 1}{d^2(\mathbf{z}, \mathbf{p}_j) + \epsilon}\right) \tag{4}$$

By upsampling these similarity scores and weighting them by the final classification layer, we reconstruct the exact spatial map the network used to make its decision.

III. Evaluation Metrics

To ensure a fair and rigorous comparison, we implemented a custom XAI metric suite based on foundational explanation validation techniques [8]:

- **Infidelity**: Measures how the model's prediction changes when important pixels (according to explanation Φ) are perturbed by a noise vector I. Lower is better.

$$INFD = \mathbb{E}_I\left[\left(f(x) - f(x-I) - I^T\Phi(x)\right)^2\right] \tag{5}$$

- **Complexity**: Measures the sparsity of the explanation using Shannon entropy of the normalized attribution magnitudes $p_i = |\Phi_i| / \sum |\Phi_j|$. Lower is better.

$$C(\Phi) = -\sum_i p_i \ln(p_i) \tag{6}$$

- **Localization Accuracy**: Evaluated via Intersection over Union (IoU) between the thresholded explanation mask M and the ground-truth object bounding box B. Higher is better.

$$Loc = \frac{|M \cap B|}{|M \cup B|} \tag{7}$$

- **Rank Agreement (Spearman)**: Measures the correlation between the rankings of pixel importance given by two different methods, where d_i is the difference in ranks.

$$\rho = 1 - \frac{6\sum d_i^2}{n(n^2-1)} \tag{8}$$

IV. Experimental Results

A. Post-Hoc Methods on Classical Backbones

When evaluating ResNet50 and ViT, the post-hoc methods exhibited strong dependency on the dataset characteristics (see Table I).

- **MNMath (Fine-grained)**: SHAP significantly outperformed GradCAM. GradCAM's low spatial resolution (e.g., 7×7 feature maps) caused its heatmaps to blur over the fine strokes of the digits, resulting in poor localization. LIME also struggled to isolate thin strokes, as its superpixels tend to group the white background with the fine handwritten details.
- **CLE4EVR (Large Objects)**: GradCAM rebounded, performing on par with SHAP. The large 3D objects perfectly matched the scale of GradCAM's receptive field. LIME severely underperformed due to its default SLIC superpixel segmentation, which fragmented the smooth 3D objects.

TABLE I
Post-Hoc Metrics on ResNet50 and ViT (Mean over batches)

Dataset	Model	Explainer	Infidelity ↓	Complexity ↓
CLE4EVR	ResNet50	SHAP	**5.158**	**0.007**
		GradCAM	6.021	0.453
		LIME	22.782	0.831
	ViT	SHAP	**0.068**	**0.002**
		Rollout	0.134	0.087
		LIME	17.745	0.498
MNMath	ResNet50	SHAP	1.576	**0.026**
		GradCAM	**1.333**	0.893
		LIME	1.859	0.321
	ViT	SHAP	**0.225**	**0.029**
		Rollout	0.228	1.000
		LIME	2.656	0.321

1) LIME Ablation Study: To investigate LIME's failure on CLE4EVR, we conducted an ablation study where LIME's segment count was forced to match the expected number of objects in the scene. While visually improving the superpixel alignment to object boundaries, it drastically increased infidelity (see Table II), suggesting that linear surrogates struggle with highly non-linear spatial reasoning over large objects.

TABLE II
LIME Ablation on CLE4EVR: Superpixel Count vs. Infidelity

Superpixels (K)	Alignment Quality	Infidelity ↓
Default (50)	Fragmented	16.010
Match Objects (≈ 4)	Object-Aligned	34.250
Dense (100+)	Highly Fragmented	12.100

B. Post-Hoc vs. Intrinsic Interpretability

The most significant finding of this study emerged when evaluating ProtoPNet. We ran SHAP and LIME on ProtoPNet and compared them to ProtoPNet's internal similarity heatmaps (see Table III).

As shown by the negative rank correlation (-0.09 on CLE4EVR, -0.24 on MNMath) in Fig. 2, SHAP and the internal reasoning of ProtoPNet are in complete disagreement.

- **SHAP** highlights fine edges and bright pixels, producing a highly sparse explanation (**Complexity: 0.001**).
- **ProtoPNet Internal** highlights broad, diffuse spatial regions, including the objects and their surrounding context (**Complexity: 0.52**, see Fig. 3).

While SHAP achieves higher localization accuracy against strict bounding boxes, it fails to represent how the network

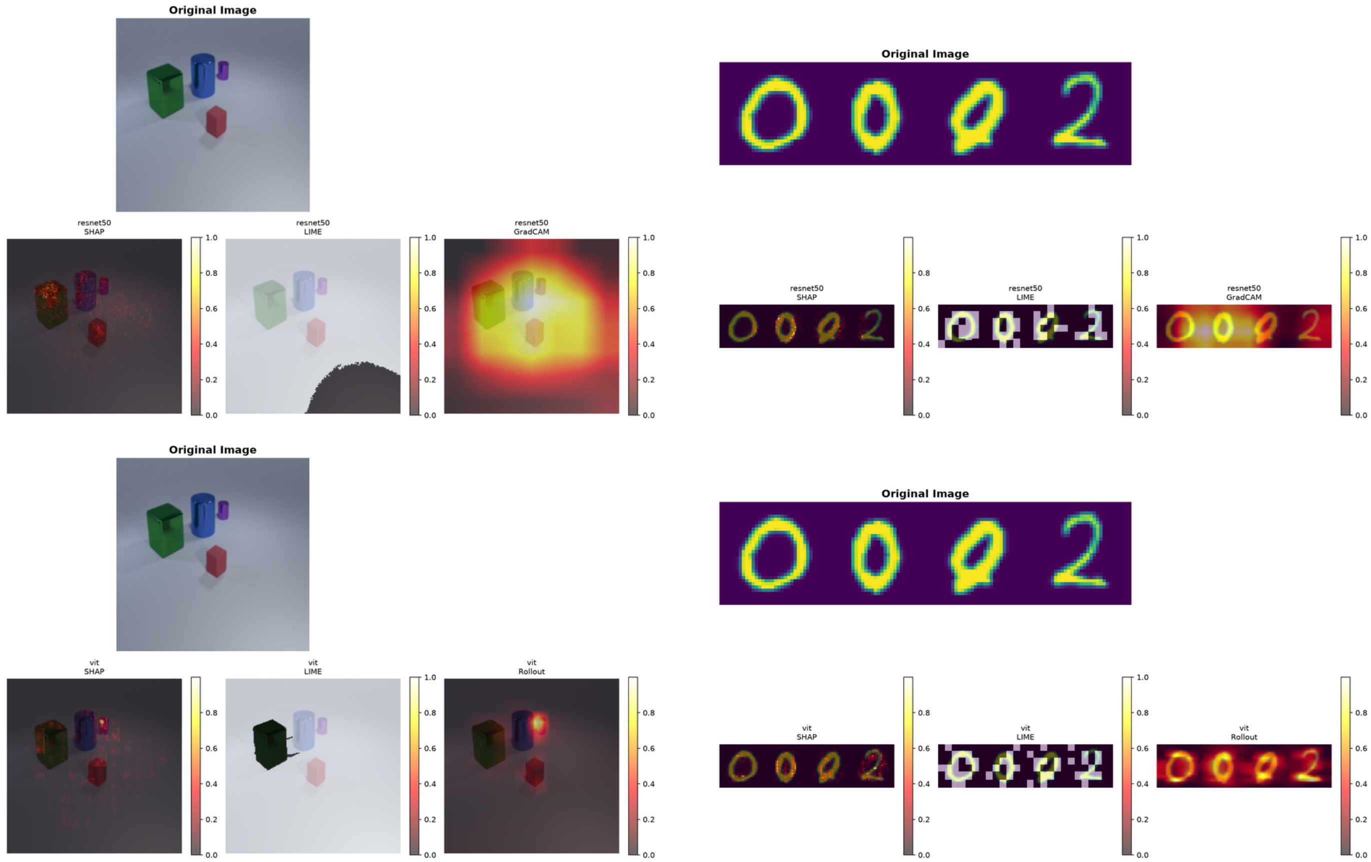

Fig. 1. Visual comparison of post-hoc methods. Top Left: ResNet50 on CLE4EVR. Top Right: ResNet50 on MNMath. Bottom Left: ViT on CLE4EVR. Bottom Right: ViT on MNMath.

TABLE III
PROTOPNET INTRINSIC VS POST-HOC METRICS

Dataset	Method	Infidelity ↓	Complexity ↓	Localization ↑
CLE4EVR	SHAP	0.130	**0.001**	0.184
	LIME	16.010	0.272	0.106
	Internal	9.081	**0.523**	0.084
MNMath	SHAP	46.544	**0.008**	0.516
	LIME	44.774	0.318	0.271
	Internal	46.908	**0.170**	0.173

actually reasons. ProtoPNet learns prototypes that capture broader contextual relationships (e.g., ”object glow” or ”inter-object spacing”) rather than strict object boundaries. Figure 4 provides a visual representation of these learned prototypes.

V. CONCLUSION

This study demonstrates the risks of relying solely on post-hoc explainers. Methods like SHAP and LIME provide neat, edge-aligned visual explanations that align with human intuition, but they fundamentally disagree with the actual internal cognitive mechanisms of models designed for intrinsic interpretability [1]. Evaluating XAI methods strictly on ”localization accuracy” can be misleading if the ground truth masks assume networks reason like humans. Future work will extend this evaluation framework to hybrid quantum-classical neural networks (Hybrid QCNN/QViT).

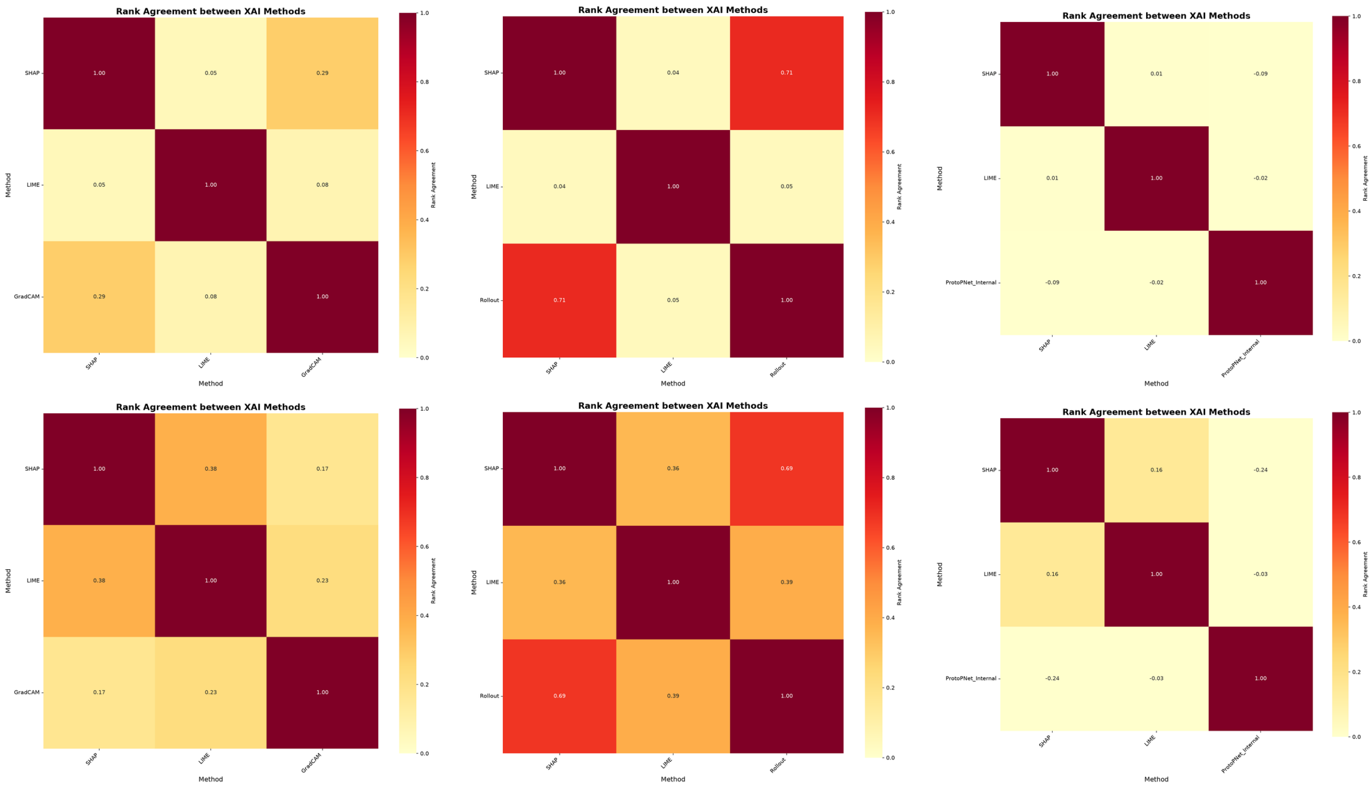

Fig. 2. Rank Agreement Heatmaps comparing Spatial Locality and Infidelity. Top row: CLE4EVR. Bottom row: MNMath. Columns from left to right: ResNet50, ViT, ProtoPNet.

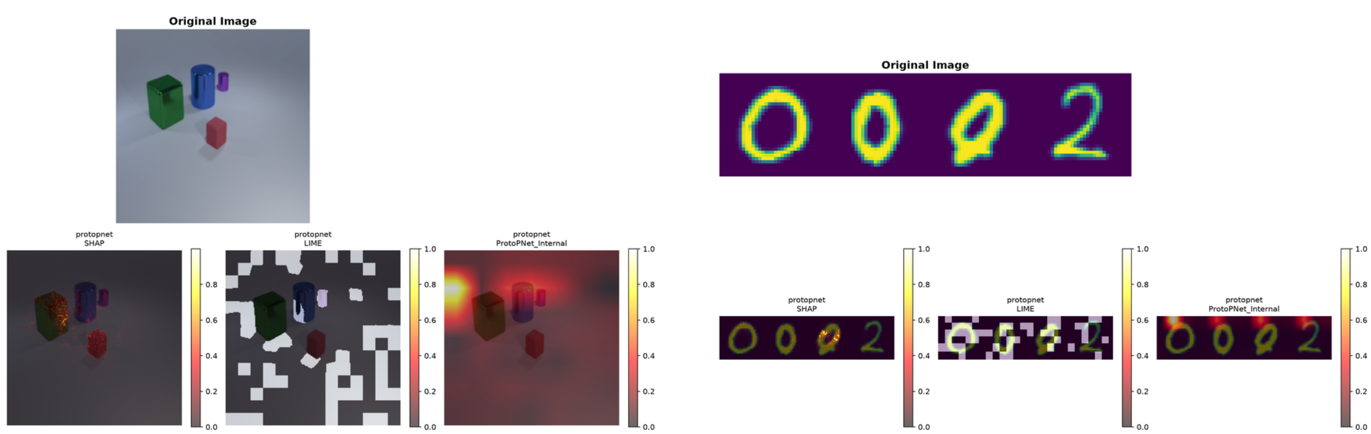

Fig. 3. Visual comparison of ProtoPNet Internal vs Post-Hoc. Left: CLE4EVR. Right: MNMath. ProtoPNet captures broad contexts, contradicting post-hoc methods.

Fig. 4. Visual representation of learned prototypes by ProtoPNet. Left: Examples of prototypes on CLE4EVR. Right: Examples on MNMath.

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